

Predicting Aerodynamic Pressure Fields on Cars with Gauge Equivariant Mesh CNNs

Replacing hours of CFD simulation with seconds of neural network inference



GEOMETRIC DEEP LEARNING × CFD

DRVAERNET DATASET

TRAINED ON 200 F_S_WWS_WM MESHES

The Problem with CFD

Traditional CFD is Slow

- Navier-Stokes equations on millions of mesh cells
- **2-8 hours** per design iteration
- Requires HPC cluster
- Cannot support rapid design exploration



Design iteration speed is the bottleneck in modern aerodynamic engineering.

The GEM-CNN Surrogate

Predictions within seconds,
once trained

Per design — vs. 2–8 hours for full
CFD

Used 50K Vertices

Single GPU inference, no HPC
required

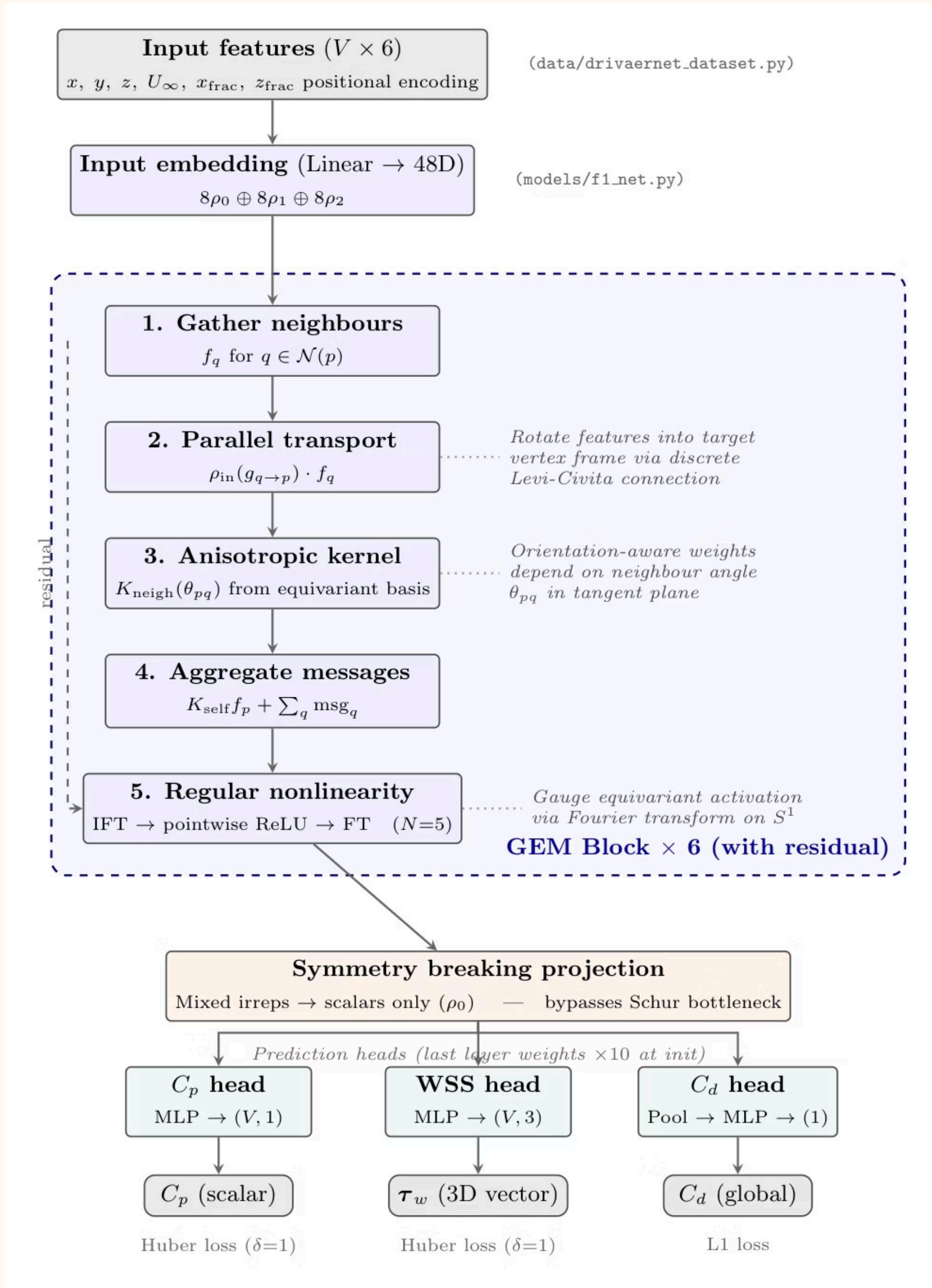
Geometry-Aware

Learns directly from mesh structure

Key insight: Standard GCNs use **isotropic** filters — they cannot distinguish a leading edge from a trailing edge. Gauge equivariant convolutions solve this.

✅ GEM-CNN could be used to test prototype meshes with high accuracy and low computational cost

The Architecture



de Haan, Weiler, Cohen & Welling (2020)

Anisotropic Kernel

Kernel value depends on angular position θ_{pq} of neighbour in the tangent plane — direction matters.

Parallel Transport

Rotates neighbour features into the target vertex frame before aggregation — essential on curved surfaces.

SO(2) Irreps

Scalars (ρ_0), tangent vectors (ρ_1), tensors (ρ_2) each transform correctly under gauge change.

The Heart of Gauge Equivariance

The core mathematical formulation enabling geometry-aware learning on meshes.

Key equations

GEM convolution

$$(K \star f)_p = K_{\text{self}} f_p + \sum_{q \in \mathcal{N}(p)} K_{\text{neigh}}(\theta_{pq}) \rho(g_{q \rightarrow p}) f_q$$

SO(2) irreducible representations

$$\rho_0(g) = 1 \quad (\text{scalars: } C_p, \text{ invariant}) \quad \rho_n(g) = \begin{pmatrix} \cos ng & -\sin ng \\ \sin ng & \cos ng \end{pmatrix} \quad (\text{vectors, tensors})$$

Kernel equivariance constraint

$$K(\theta - g) = \rho_{\text{out}}(-g) K(\theta) \rho_{\text{in}}(g) \quad \forall g \in \text{SO}(2)$$

Symlog transform

$$\text{symlog}(x) = \text{sign}(x) \cdot \ln(1 + |x|)$$

Huber loss ($\delta = \frac{1}{2}$)

$$L_\delta(e) = \begin{cases} \frac{1}{2}e^2 & |e| \leq \delta \\ \delta(|e| - \frac{\delta}{2}) & |e| > \delta \end{cases}$$

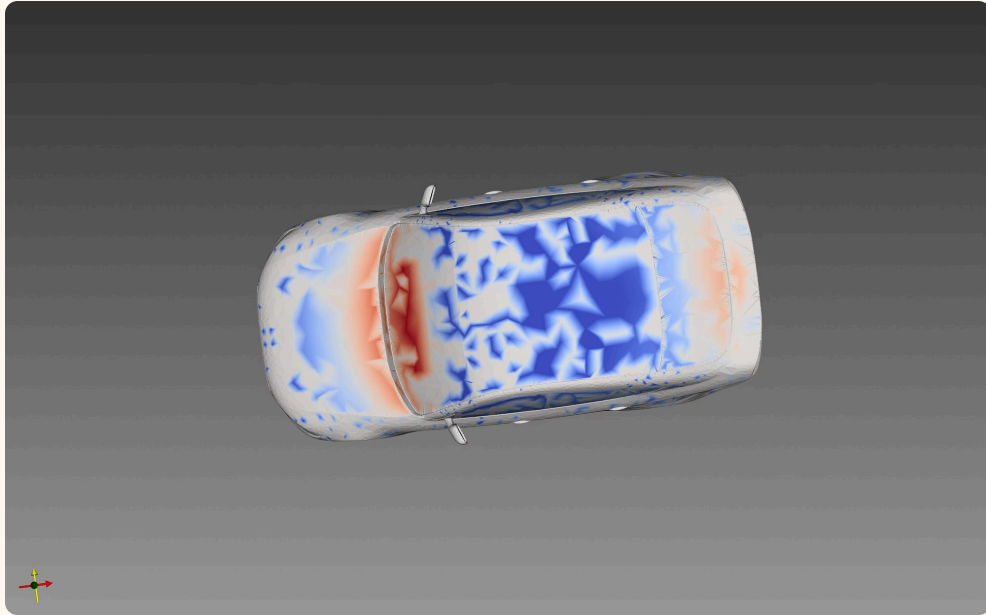
Layer feature types

$$8\rho_0 \oplus 8\rho_1 \oplus 8\rho_2 \rightarrow 16\rho_0 + 16\rho_1 + 16\rho_2 \rightarrow 32\rho_0 \oplus 32\rho_1 \rightarrow 8\rho_1 \oplus 8\rho_1 \rightarrow \text{heads}$$

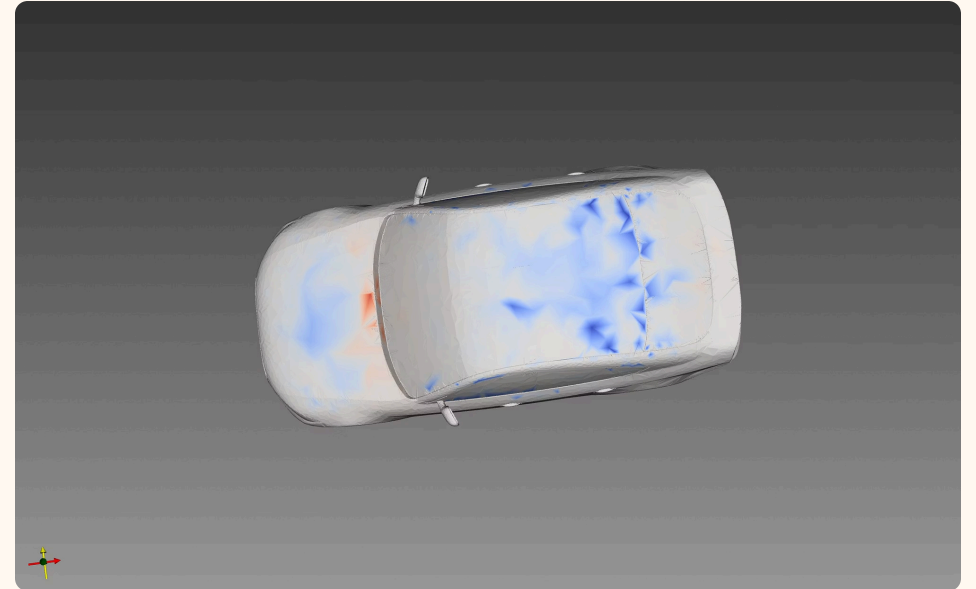
i g is the group element and $\rho_n(g)$ represents the irrep of g in the respective feature space

(Source: de Haan, Weiler, Cohen & Welling, 2020)

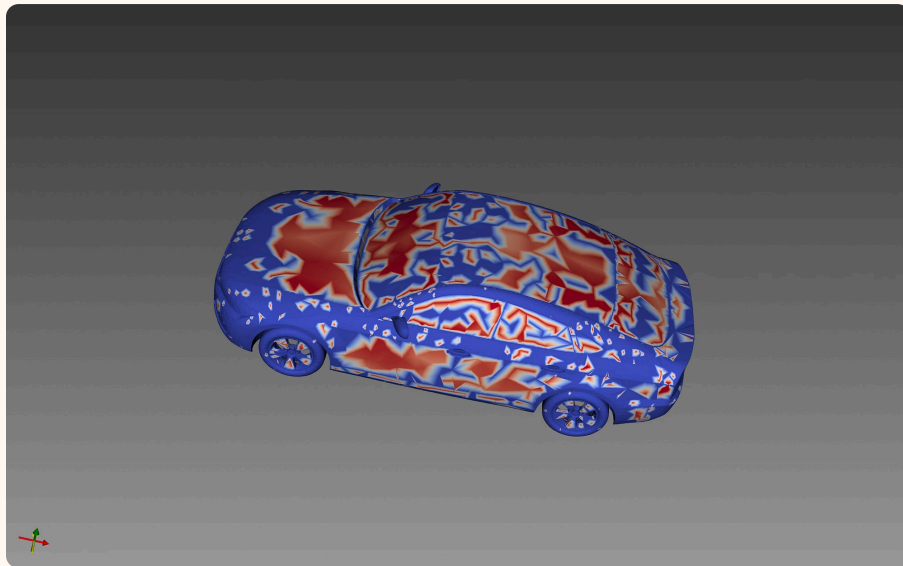
Paraview outputs



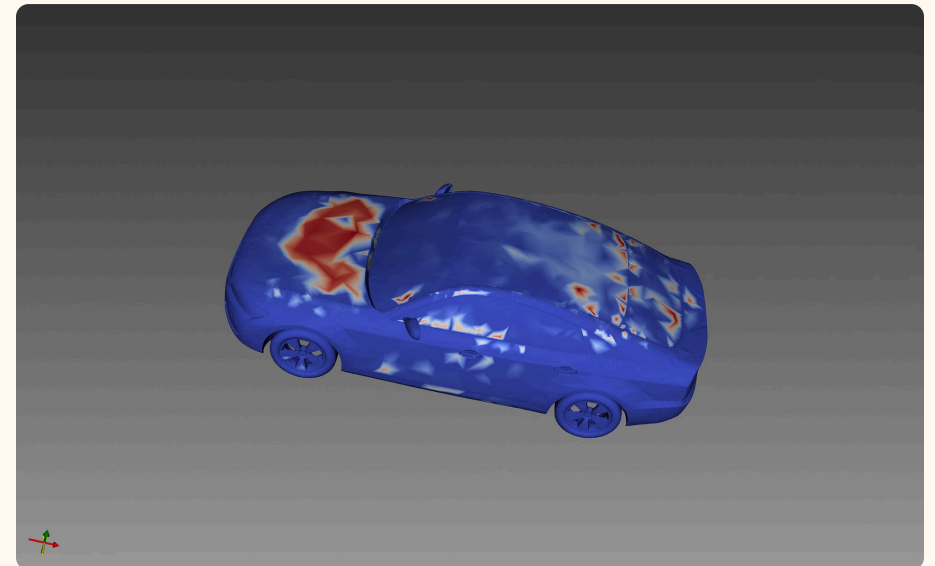
Cp True



Cp Predicted

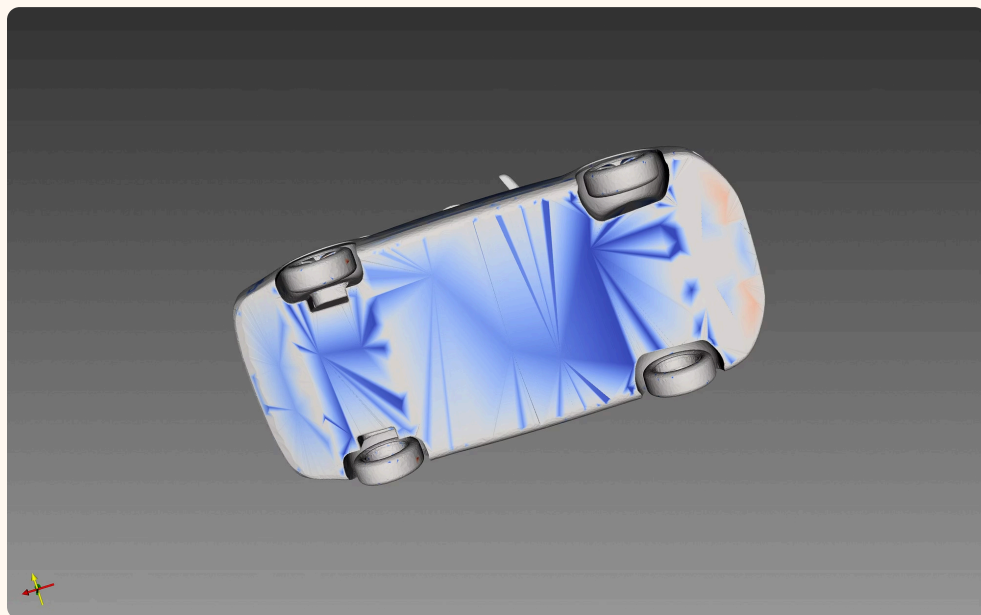


WSS True

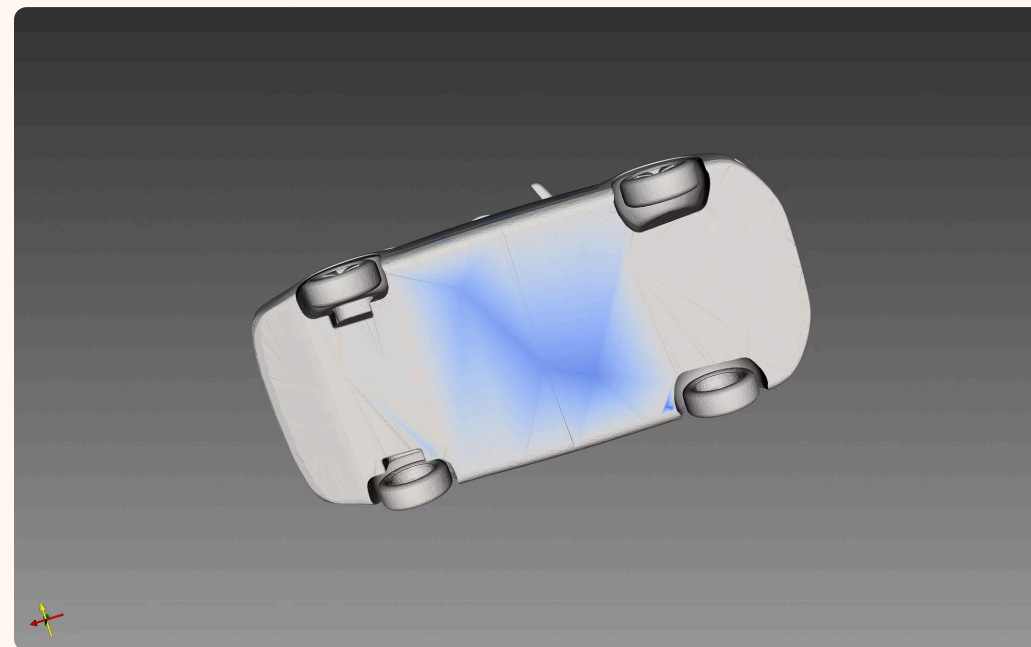


WSS Predicted

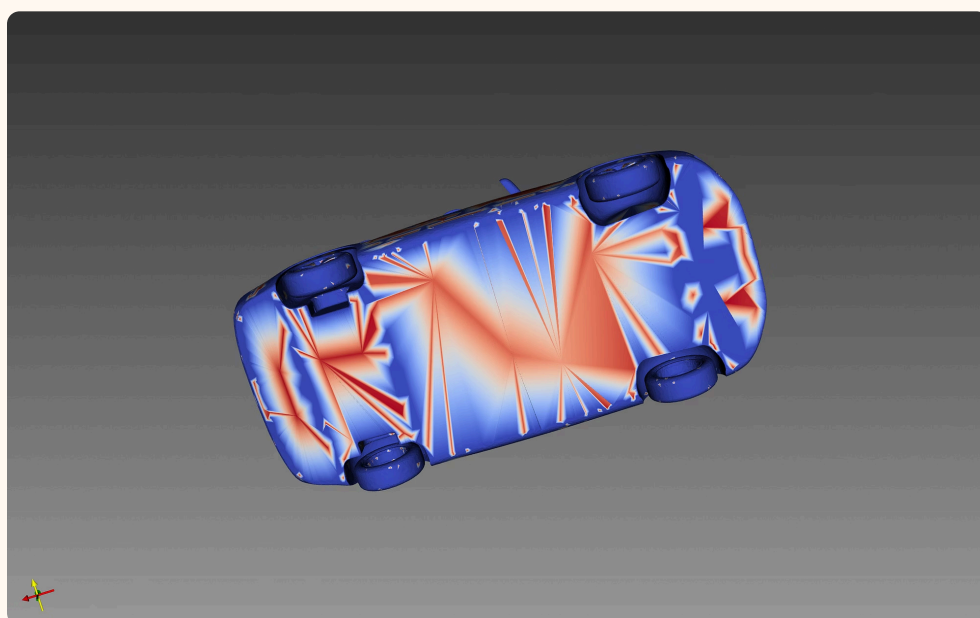
Cp True



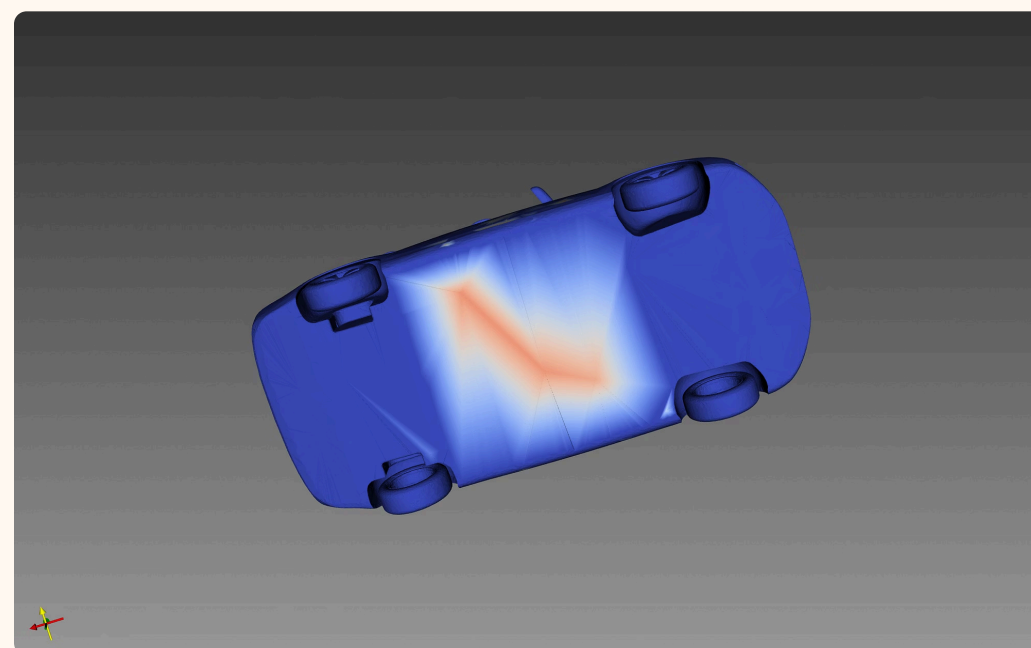
Cp Predicted



WSS True



WSS Predicted



What Broke — And How It Was Fixed

Loss was stuck at **2.0 for weeks**. Three compounding bugs:

Bug 1: Output Scale Mismatch

MLP heads output std ≈ 0.05 vs. target std = 1.0. **Fix:** Multiply last-layer weights by 10 $\times$ at init.

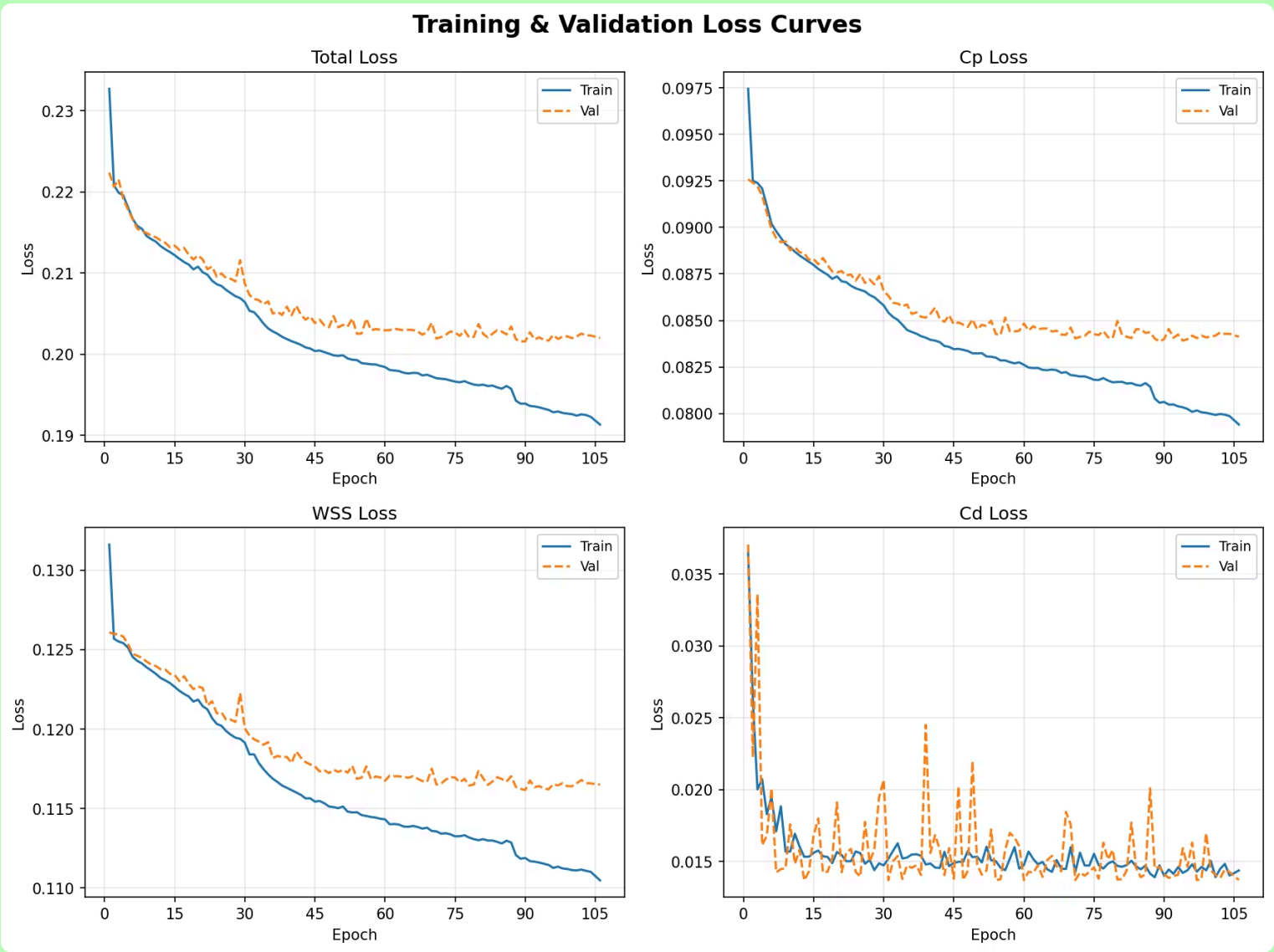
Bug 2: Heavy-Tailed CFD Data

Pressure spikes at $\pm 27\sigma$ dominated MSE gradients. **Fix:** symlog transform

Bug 3: Model converged to the average

Because of big values at the tail, the model played safe and predicted the mean value for all points in case of MSE. **Fix :** Huber loss ($\delta=1.0$) + $\pm 5\sigma$ clipping.

Training vs Validation Loss



Loss dropped from **2.0** → **0.19** after all three fixes were applied.

0.19

Final Loss

Down from 2.0

Current Limitations

Receptive Field

≈12 vertices (6 GEM layers) — fine-scale features not yet resolved

No Global Context

Needs mesh pooling for long-range aerodynamic interactions

Single Config

Trained on one body style only

Low angle variations

Max rep allowed in the model is ρ_2 , which doesn't allow the model to pick higher angular variations.

Next Steps

Wider architecture

Mesh pooling (U-Net style)

More Kernel parameters

Multi-configuration : fastback + estateback+ notchback

Let's Connect

Built with **PyTorch** · **PyG** · **GEM-CNN** · **DrivAerNet++** · **ParaView** · **Kaggle T4**

Geometric Deep Learning

Gauge equivariant convolutions on arbitrary meshes

CFD Surrogate Modeling

Seconds vs. hours — enabling real-time design iteration

Open to Collaborate

Reach out if you want to collab on GDL or TDA for scientific ML